

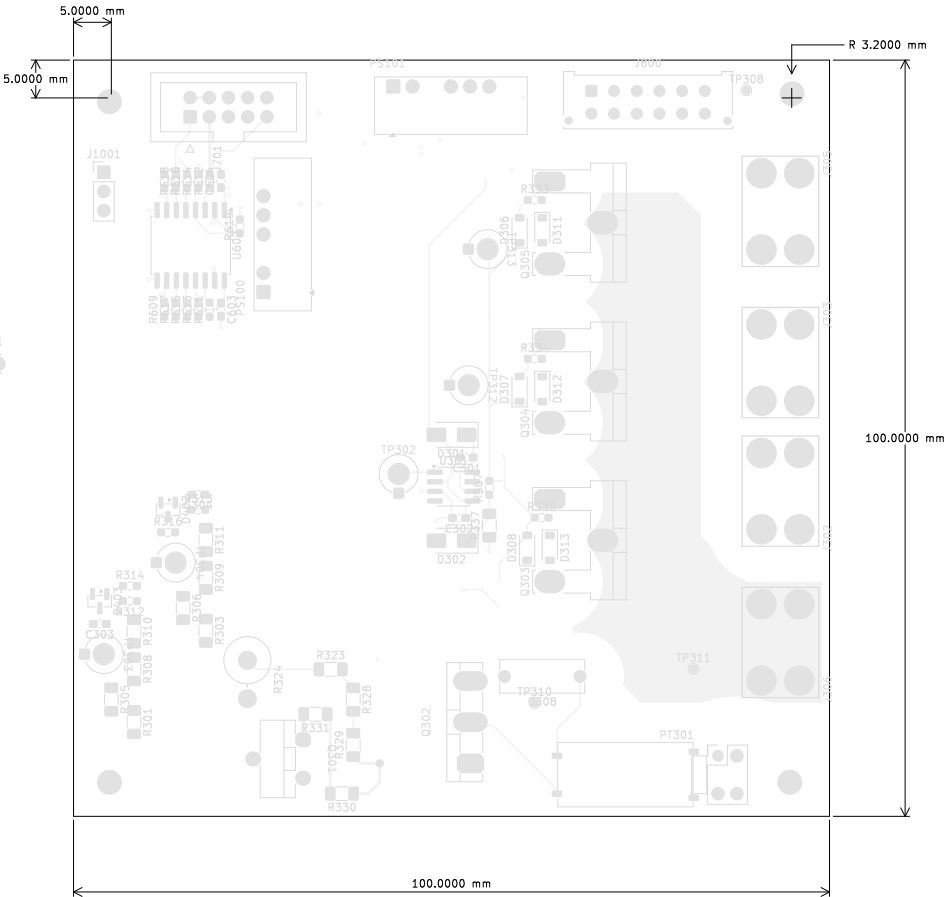
Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

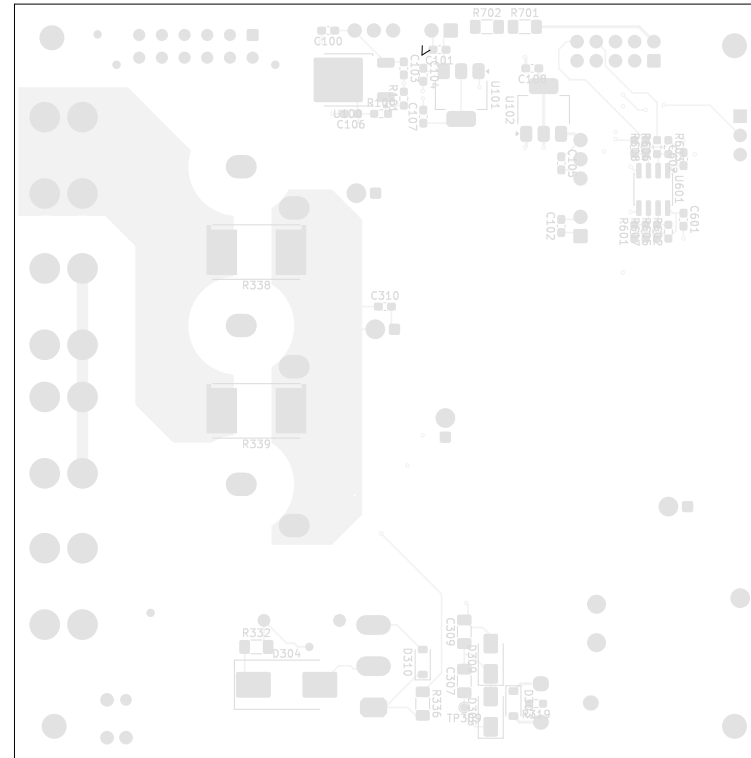
Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	
F.Silkscreen				Legend	GBR
F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
Copper	F.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	In1.Cu (GND)	0.035mm (1.00oz)		Plane	GBR
Core		1.828mm	FR4	Dielectric	
Copper	In2.Cu (Sig, PWR)	0.035mm (1.00oz)		Plane	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	B.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
B.Silkscreen				Legend	GBR
B.Paste				Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating



Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



1 kV strobe
04/Jul/25
Felipe L Rodrigues



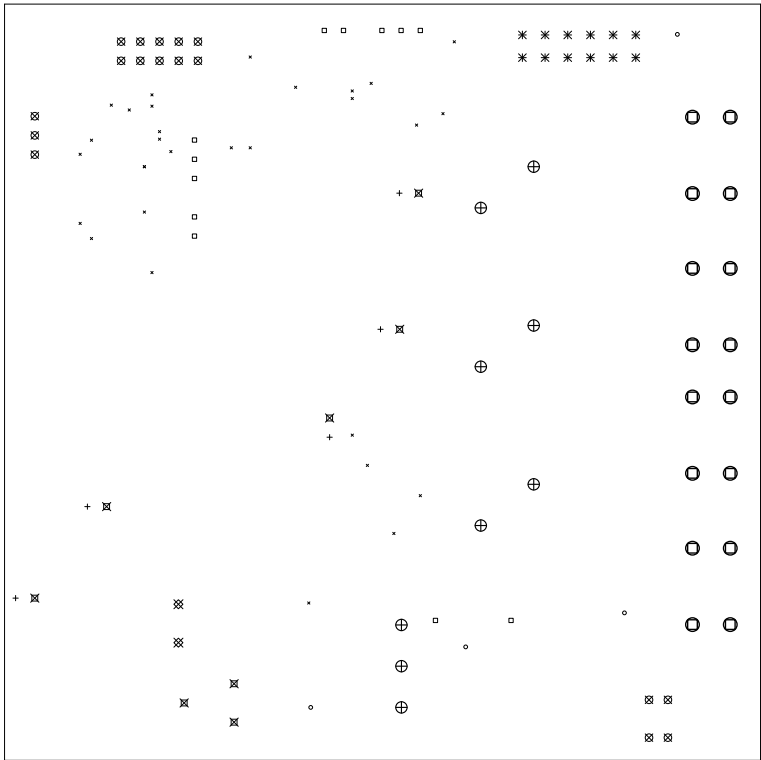
Comments:	Company: CurrasCod Electronics		Variant: PRELIMINARY	Git Hash: f657b35
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 2 of 10

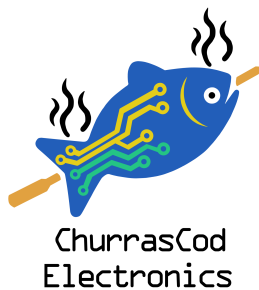
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	30	0,30mm (11,81mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
○	24	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
+	5	0,70mm (27,56mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
□	12	0,80mm (31,50mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◇	2	0,90mm (35,43mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	20	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
※	12	1,02mm (40,16mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊠	8	1,10mm (43,31mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	2	1,20mm (47,24mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	9	1,50mm (59,06mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	16	1,80mm (70,87mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
Total 140						



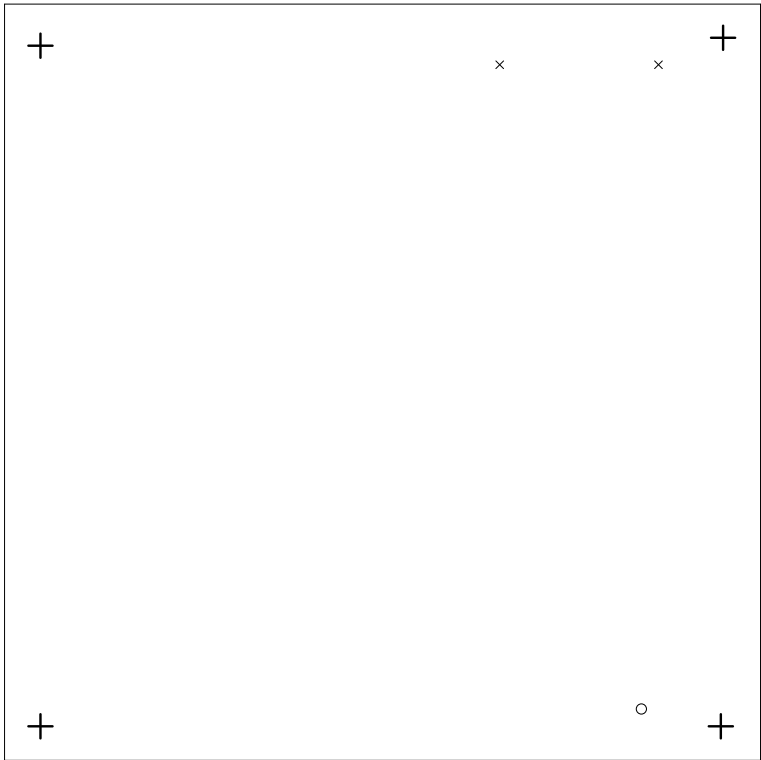
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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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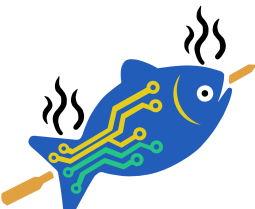
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1,02mm (40,16mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
○	1	1,35mm (53,15mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
	Total 7					



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Main board Fabrication Document

Top Test Points (Scale 1:1)

A

B

C

D

E

F

A

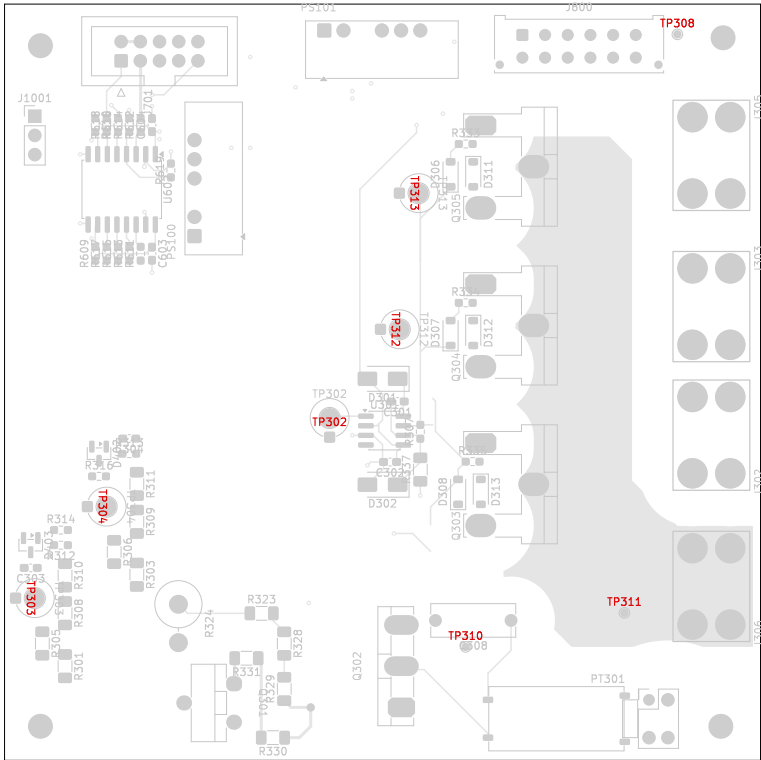
B

C

D

E

F



Ref.	Net	X [mm]	Y [mm]
TP302	SH0	48.00	49.00
TP303	SH0	7.73	26.45
TP304	Mon_S1k	17.23	38.55
TP308	S1k	94.00	101.00
TP310	TP_PRI	66.00	20.00
TP311	LMP	87.00	24.50
TP312	SH0	56.00	62.00
TP313	SH0	58.50	80.00
TP501	Net-(U501-PA0)	-28.90	71.78
TP502	Net-(U501-PA1)	-28.90	68.73
TP503	SUP_TX	-28.90	65.68
TP504	SUP_RX	-28.90	62.63
TP505	Mon_S1k_buf	-25.86	71.78
TP506	Net-(U501-PB0)	-25.86	68.73
TP507	Mon_IL_prot	-25.86	65.68
TP508	Net-(U501-PB3)	-25.86	62.63
TP509	Mon_LMP_buf	-22.80	71.78
TP510	Net-(U501-PB4)	-22.80	68.73
TP511	Net-(U501-PA7)	-22.80	65.68
TP512	Net-(U501-PB5)	-22.80	62.63
TP513	Vbus_ctrl	-19.75	71.78
TP514	Light_sns	-19.75	68.73
TP515	INT_EN	-19.75	65.68
TP516	Net-(U501-PA12)	-19.75	62.63
TP517	Net-(U501-PA15)	-16.70	71.78
TP518	BOOT0	-16.70	68.73
TP519	SH0	-16.70	65.68
TP520	SH0	-16.70	62.63

Lim. resistor
PFC

HMI

Ext
ctrl

SWD
Status

1 kV strobe
04/Jul/2025

Boot

Light
sense

Felipe L Rodrigues

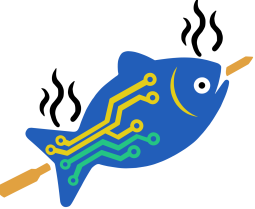
Cap+

Cap-

Lamp
trig

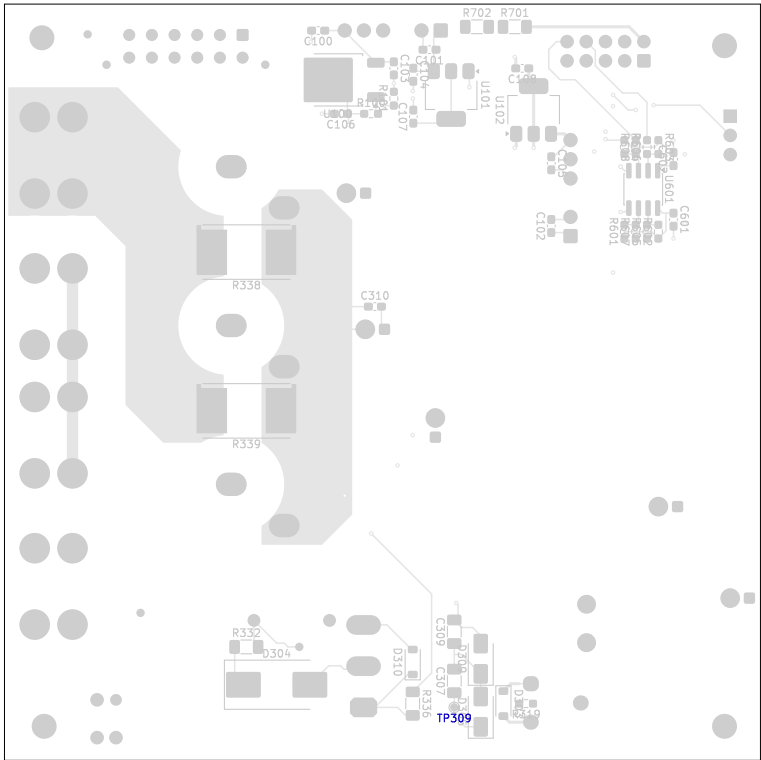
Lamp+

Lamp-

 ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: f657b35
	Sheet Title: Top Test Points (Scale 1:1)	Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Path:	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
			Reviewer:	Size: A3	Sheet: 5 of 10

Main board Fabrication Document

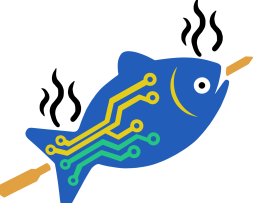
Bottom Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP309	TP_REF	45.50	12.00

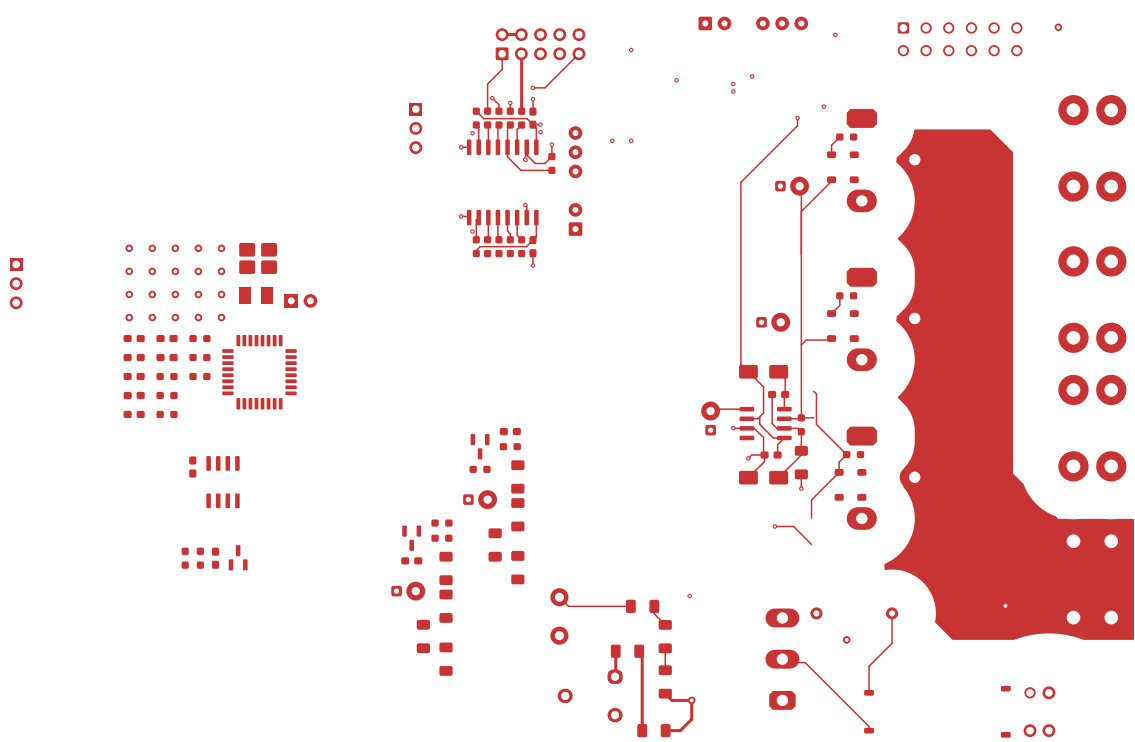


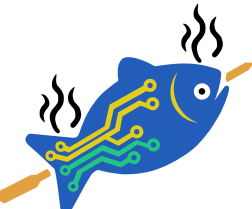
1 kV strobe
04/Jul/25
Felipe L. Rodrigues

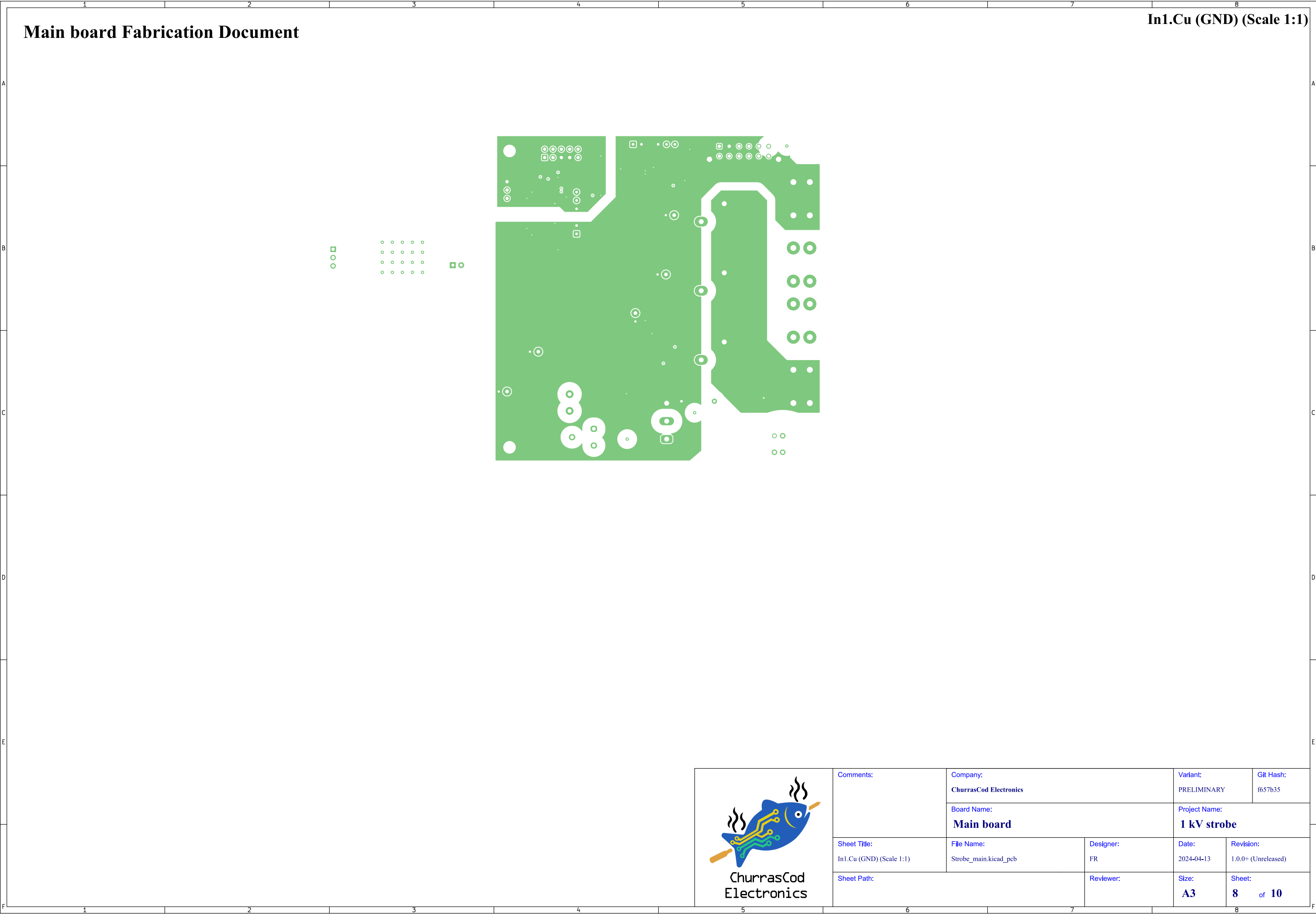
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		Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:			Reviewer:	Size: A3	Sheet: 6 of 10

Main board Fabrication Document

F.Cu (Sig, PWR) (Scale 1:1)

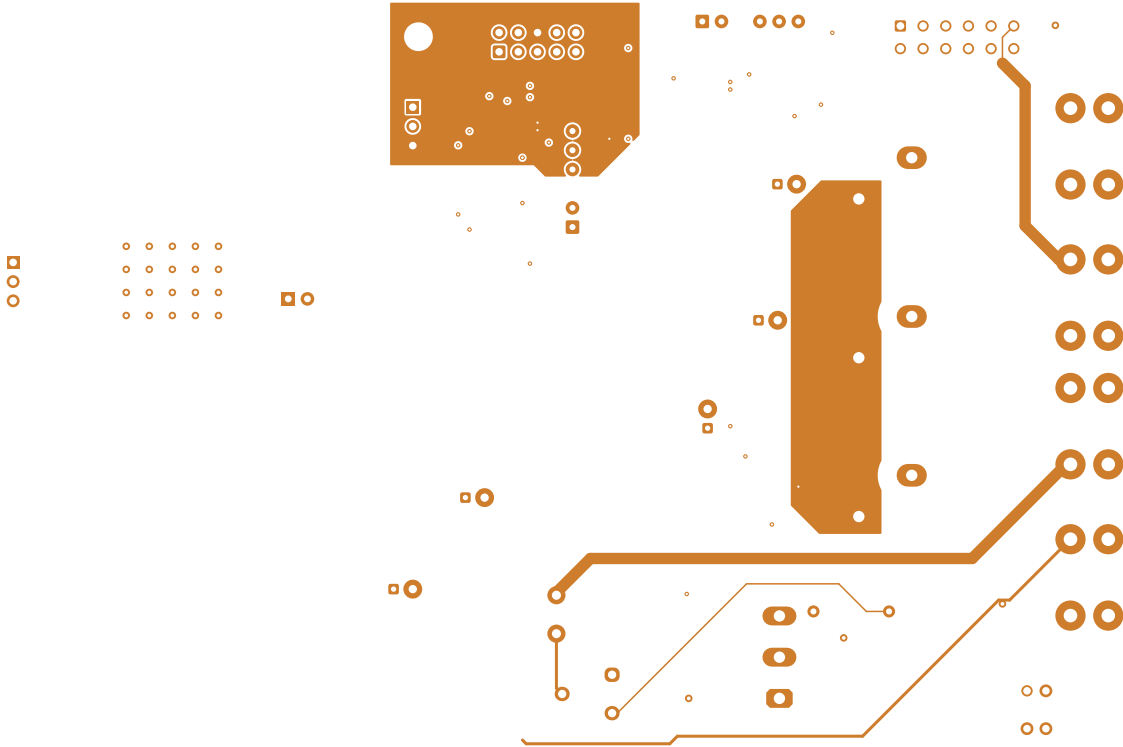


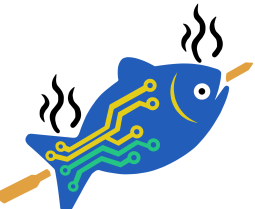
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	Sheet Title: F.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Main board Fabrication Document

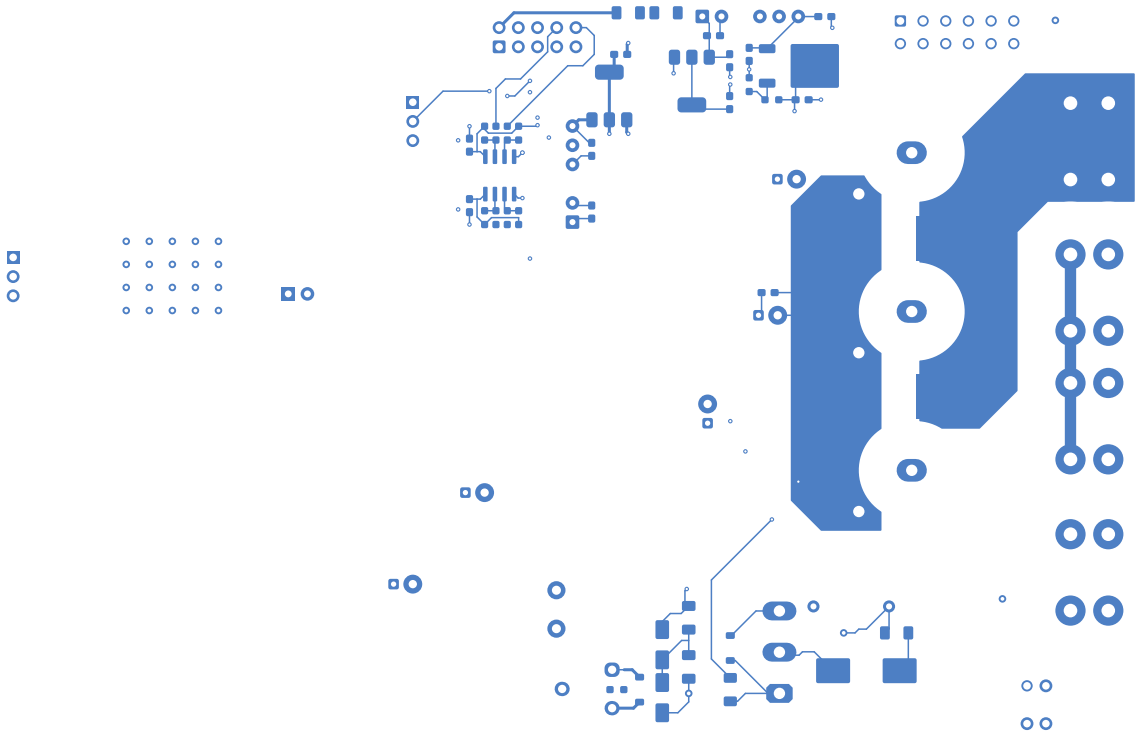
In2.Cu (Sig, PWR) (Scale 1:1)

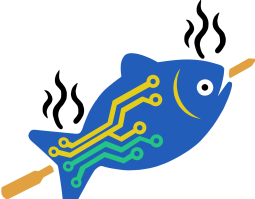


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	Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:	Reviewer:		Size: A3	Sheet: 9 of 10

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY		Git Hash: f657b35			
			Board Name: Main board		Project Name: 1 kV strobe					
	Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)		File Name: Strobe_main.kicad_pcb		Designer: FR		Date: 2024-04-13		Revision: 1.0.0+ (Unreleased)	
	Sheet Path:				Reviewer:		Size: A3		Sheet: 10 of 10	